$$\int \frac{3x^2 - x + 8}{x^3 + 4x}\, dx$$

$$x(x^2 + 4)$$

$$\frac{3x^2 - x + 8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

$$3x^2 - x + 8 = A(x^2 + 4) + (Bx + C)x$$

$x = 0 \qquad 8 = 4A \qquad A = 2$

$$3x^2 - x + 8 = 2x^2 + 8 + Bx^2 + Cx$$

Match corresponding terms on each side

$3x^2 = 2x^2 + Bx^2 \qquad B = 1$

$-x = Cx \qquad C = -1$

$$\frac{3x^2 - x + 8}{x(x^2+4)} = \frac{2}{x} + \frac{x-1}{x^2+4}$$

$$\int \frac{3x^2 - x + 8}{x(x^2+4)}\,dx = \int \left(\frac{2}{x} + \frac{x}{x^2+4} - \frac{1}{x^2+4} \right) dx$$

$$= 2\ln|x| + \frac{1}{2}\ln|x^2+4| - \frac{1}{2}\tan^{-1}\frac{x}{2} + C$$

$$\int \frac{x}{x^2+4}\,dx$$

$u = x^2+4$

$du = 2x\,dx$

$\frac{1}{2}du = x\,dx$

$$\int \frac{1}{u}\cdot\frac{1}{2}\,du$$

$$\frac{1}{2}\ln|u| = \frac{1}{2}\ln|x^2+4|$$

$$\int \frac{1}{x^2+4}\,dx$$

$x = 2\tan\theta$

$dx = 2\sec^2\theta\,d\theta$

$x^2+4 = 4\tan^2\theta+4$

$= 4\sec^2\theta$

$$\int \frac{1}{4\sec^2\theta}\cdot 2\sec^2\theta\,d\theta$$

$\int \frac{1}{2}\, d\theta$

$x = 2\tan\theta$

$\tan^{-1}\frac{x}{2} = \theta$

$\frac{1}{2}\theta$

$\frac{1}{2}\tan^{-1}\frac{x}{2}$

Fact you can use:

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

$$\int \frac{x+4}{x^2+2x+6}\,dx$$

$2^2 - 4\cdot 6 < 0$

divide by 2

$$(x+1)^2+5$$

x^2+2x+1

$$\int \frac{x+4}{(x+1)^2+5}\,dx$$

$u = x+1$ $\quad x = u-1$

$du = dx$

$$\frac{3}{u^2+5} \qquad \frac{3}{u^2+(\sqrt{5})^2}$$

$$\int \frac{u+3}{u^2+5}\,du = \int \left(\frac{u}{u^2+5} + \frac{3}{u^2+5}\right) du$$

$$\frac{1}{2}\ln|u^2+5| + 3\cdot\frac{1}{\sqrt{5}}\tan^{-1}\frac{u}{\sqrt{5}} + C$$

$$\int \frac{u}{u^2+5}\,du \qquad v = u^2+5$$

$$dv = 2u\,du$$

$$\int \frac{1}{v}\,\frac{1}{2}\,dv$$

$$\frac{1}{2}\ln|v| = \frac{1}{2}\ln|u^2+5|$$

$$\frac{1}{2}\ln|u^2+5| + 3\cdot\frac{1}{\sqrt{5}}\tan^{-1}\frac{u}{\sqrt{5}} + C \qquad u = x+1$$

$$\frac{1}{2}\ln|x^2+2x+6| + \frac{3}{\sqrt{5}}\tan^{-1}\left(\frac{x+1}{\sqrt{5}}\right) + C$$

$$\int \frac{1}{2\sqrt{x+3}+x}\,dx$$

$u = \sqrt{x+3}$

$du = \frac{1}{2\sqrt{x+3}}\,dx$

$$\int \frac{1}{2u+(u^2-3)}\,2u\,du$$

$2\sqrt{x+3}\,du = dx$

$2u\,du = dx \qquad x = u^2-3$

$$\int \frac{2u}{u^2+2u-3}\,du$$

$(u+3)(u-1)$

$$\frac{2u}{(u+3)(u-1)} = \frac{A}{u+3} + \frac{B}{u-1}$$

$$2u = A(u-1) + B(u+3)$$

$u=1 \quad 2 = 4B \quad B = \frac{1}{2}$

$u=-3 \quad -6 = -4A \quad A = \frac{3}{2}$

$$\int \frac{2u}{(u+3)(u-1)}\,du = \int \left(\frac{3/2}{u+3} + \frac{1/2}{u-1}\right) du$$

$$= \frac{3}{2}\ln|u+3| + \frac{1}{2}\ln|u-1| + C$$

$$= \frac{3}{2}\ln|\sqrt{x+3}+3| + \frac{1}{2}\ln|\sqrt{x+3}-1| + C$$

$e^x \quad \sin x \quad \cos x \quad e^{-x}$

Strategy for Integration (7.5)

Recall from the Fundamental Theorem of Calculus pt 1 that every continuous function $f(x)$ has an antiderivative, namely

$$F(x) = \int_a^x f(t)\,dt$$

for any a in the domain of f.

However, most functions do not have elementary antiderivatives, i.e., the function F above usually cannot be written in terms of polynomials, exponentials, trig functions, etc.

Common real-world examples of nonelementary integrals:

$$\int e^{-x^2}\,dx$$



$$\int \frac{e^x}{x}\,dx \qquad \int \frac{\sin x}{x}\,dx$$

$$\int \frac{1}{\ln x}\,dx$$

$$\int \sqrt{x^3+1}\,dx$$

elliptic integral

$$\int \sin(x^2)\,dx \qquad \int \sin(e^x)\,dx \qquad \int e^{e^x}\,dx$$

In this class, if you are asked to evaluate an integral, you can be sure that you can do it, and in a reasonable amount of time. However, you usually won't be told which technique(s) to use.

When you practice integration in your homework, you are not only practicing the techniques, but also gaining intuition for recognizing which techniques to use.

Sometimes the first thing you try won't work, and you'll have to try something else.

Recognizable forms:

Anything on your table of integrals.

Trigonometric integrals.

$$\int \sin^m x \cos^n x \, dx \quad \text{or} \quad \int \tan^m x \sec^n x \, dx$$

m, n nonnegative integers.

Rational functions:

$$\int \frac{P(x)}{Q(x)} dx, \quad P, Q \text{ polynomials.}$$

If the integral is not immediately of a recognizable form, try one of the following:

1) Simplify / manipulate the expression.

2) Substitution

- u-sub (see next page)
- trig sub (if you see $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, $\sqrt{x^2-a^2}$)

3) Integrate by parts (good for $x^n \cdot (\ldots)$ where $(\ldots)$ = e^x, $\sin x$, $\cos x$)

$(\ln x)$ x^n u dv

U-sub tips.

- Try to look for u such that du also appears in the integral, e.g. $\int \frac{x^2}{\sqrt{x^3+1}}\,dx$

- Substitutions of the form $u=\sqrt{ax+b}$ often work well because

$$du = \frac{a}{2\sqrt{ax+b}}\,dx \qquad x = \frac{1}{a}(u^2-b)$$

$$2\sqrt{ax+b}\,du = a\,dx$$

$$2u\,du = a\,dx$$